A Novel Algorithm for Leaf Incidence Angle Effect Correction of Hyperspectral LiDAR

Jie Bai, Shuai Gao, Zheng Niu, Changsai Zhang, Kaiyi Bi, Gang Sun, and Yanru Huang

***Abstract*— As a novel remote sensor, hyperspectral LiDAR is faced with the incidence angle effect, which restricts its quantitative applications. However, the current radiometric correction algorithms have some limitations, concentrating on: 1) the mathematically polynomial fitting; 2) adjacent wavelength ratio such as ratio vegetation index; and 3) perfect Lambertian assumption and using the Lambert cosine law to correct the effect. In this study, a practical and proper correction algorithm is proposed to overcome these limitations. First, to better characterize the complex reflection characteristics of the object surface, a combination of the Lambert law and Beckmann law is applied to represent the object surface. Then, it considers the impact of both wavelength and incidence angle on describing the surface roughness factor and diffuse fraction. Finally, a modified and physically based radiometric correction algorithm is generated. It provides the detailed correction equations for intensity and reflectance data recorded by hyperspectral LiDAR. To obtain its parameters and verify its reliability, leaves (ten samples per species, 30 in total) were stochastically collected from three broadleaf trees to experiment. The results showed that the algorithm achieved good performances by comparing the intensity and reflectance changes before and after removing the leaf incidence angle effect. Since it is physically based, the algorithm could promisingly be a fundamental solution to eliminate the incidence angle effect for hyperspectral LiDAR.**

***Index Terms*— Hyperspectral LiDAR, incidence angle effect, radiometric correction, surface roughness factor.**

I. Introduction

As one of the novel remote sensors, hyperspectral LiDAR plays a crucial role in acquiring geometry and hyperspectral information simultaneously [1]–[3], but it is also faced with the problem of the incidence angle effect like traditional LiDAR [4]–[6]. As for one leaf blade, the backscattering intensity data recorded by hyperspectral LiDAR decrease as the incidence angle increases because of more laser energy dispersion and less returned at larger incidence angles [7]. Leaf surface is not a Lambertian reflector, and the backscattering intensity data of the leaf surface do not strictly comply with Lambert cosine law [8]. Thus, it will bring errors when the data with the incidence angle effect is used in subsequent biochemical content estimation without correcting it. One of the current main challenges of hyperspectral LiDAR is supplying intensity data with a physically based radiometric correction model that is likely to take into account the non-Lambertian characteristic of the leaf surface [9].

Incidence angle effect correction of intensity and reflectance data has been studied for merely about ten years on commercial single- or dual-wavelength terrestrial laser scanning (TLS) [7], [10], though its vital role in TLS intensity correction was put forward and discussed much earlier [11]. Previous studies about it have been carried out for single- or dual-channel LiDAR data [12], [13], but the correction models have some limitations. Ding *et al.* [12] used the Lambert law to correct the intensity by dividing the intensity at some angle by the cosine of the incidence angle. But the problem is that the object is highly possibly not a Lambertian reflector. Sanna *et al.* [13] proposed a model combining the Lommel–Seeliger law and the Lambert law to correct the intensity, but it is challenging to obtain the parameters of grain size on a microscopic scale and the projection factor in this model. Wagner [14] gave the corrected reflectance formula of a perfectly diffuse surface on LiDAR, but the main limitation of this approach is that most natural targets do not follow Lambert cosine law. A normalized reflective factor (NRF) was proposed by Qin *et al.* [15] to represent the reflection attribute of the target. However, the hypothesis is still that the natural target obeys Lambertian law, and the expression of NRF is a function of the ratio of received power to emitted power, which is also not the real definition of reflectance factor. Li *et al.* [16] introduced an empirical model for non-Lambertian reflectance calculation. But it needs first to calculate semi-ellipsoid parameters, which is very laborious, and it precludes automation of incidence angle effect correction. Xu *et al.* [17] considered leaf surface roughness and obtained corrected intensity with radiometric correction of incidence angle effect using Oren–Nayar physical model, and then pointed out that corrected intensity has a logarithmic relationship with the reflectance.

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However, the surface roughness parameter in the model is the standard deviation of slope angle distribution in radians which is intricate to retrieve. Besides, the relationship between intensity and reflectance cannot be simply expressed as an empirical logarithmic function.

For novel hyperspectral LiDAR, one more factor, wavelength, is involved in the complicated incidence angle effect. As it is a novel instrument, there are few studies on incidence angle effect correction of it. Zhang *et al.* [8] and Hu *et al.* [18] analyzed the incidence angle effect of leaf reflectance measured by indoor hyperspectral LiDAR, in which Hu *et al.* [18] built a quadratic polynomial fitting equation about the wavelength and incidence angle to calculate corrected reflectance. The formula is empirical, and it is a simple combination of variables where there is no physically based theory and not generally applicable. The incidence angle effect can be partially eliminated but not all by the ratio of the dual-channel intensities, provided that the angle effects of the two channels are similar [19]. Therefore, Zhang *et al.* [8] used some laser ratio indexes to reduce the incidence angle effect. Still, the limitation is that the effect of a single wavelength is not eliminated fundamentally, and intensity or reflectance data of a single wavelength without incidence angle effect cannot be acquired. The natural object is usually not a perfect Lambertian, and the incidence angle effect is a complicated function of reflection, wavelength, and surface roughness, in which surface roughness of the target is also a function of wavelength, cosine of incidence angle, and maximum height deviation of object's surface on the wavelength scale [20].

Incidence angle effect precludes from obtaining real reflection characteristics of leaves, affecting subsequent applications like biochemical parameter estimation [21]. The overall objective of this study is to explore a novel and physically based model to conduct incidence angle effect correction and then evaluate its performance. Precisely, it consists of: 1) establishing a quantitative and physically based correction method for the intensity and reflectance measured by hyperspectral LiDAR considering reflection, wavelength, diffuse fraction, and leaf surface roughness and 2) examining its correction reliability and robustness for different plant species with two indicators: intensity and reflectance changes before and after removing the incidence angle effect using the proposed model.

II. Materials and Methods

A. Hyperspectral LiDAR

The hyperspectral LiDAR used in this study is a prototype with a supercontinuum pulsed laser source (Fig. 1) [22], which has been applied in monitoring vegetation biochemical contents at 2-D and 3-D levels [23], [24]. The system employs a wide range of wavelengths, ranging from visible channel to near-infrared channel (409–914 nm), known to be sensitive to vegetation biochemical contents [24]. The spectral resolution of the system is 16 nm, and it can acquire 32 channels' returned echoes at each point. It has a beam divergence of 5 mrad and a receiver aperture of 80 mm. It transmits 1–2 ns pulses as the full-width at half-maximum (FWHM) of the peak at the pulse repetition rate (PRR) of 24 kHz and peak power of 20 kW.

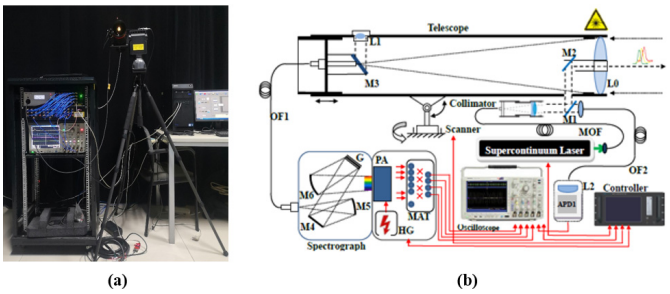


Fig. 1. Hyperspectral LiDAR prototype with a supercontinuum pulsed laser source. (a) Photograph of the prototype. (b) Optical structure of the system. L0, L1, and L2 represent the achromatic lens. M1 represents the microlens. M2 and M4 represent the reflectors. M3 represents the diffuse mirror. M5 represents the collimation mirror. M6 represents the converging mirror. G represents the grating spectrometer. APD1 represents an avalanche photo diode. MAT represents the switch matrix. OF1 and OF2 represent the optical fiber. MOF represents the microstructure optical fiber. PA represents a photosensitive array. HG represents the generator.

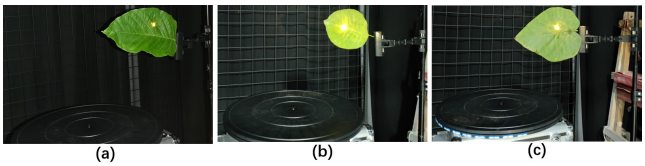


Fig. 2. Leave samples of three tree species on the compass. (a) *Magnolia delavayi.* (b) *Juglans regia.* (c) *Macaranga tanarius.*

B. Experimental Design

Leaves (ten samples per species, 30 in total) were stochastically collected from three broadleaf tree species in October in Beijing for building the correction model of the incidence angle effect and check its performance. The length of leaves ranges from 10 to 15 cm. The three tree species are *Magnolia delavayi*, *Juglans regia*, and *Macaranga tanarius.* The leaves have different surface roughness, as can be seen from Fig. 2.

To explore the incidence angle effect of backscattering intensity and reflectance, leaves and a standard white-panel with a reflectance value of 0.99 were measured at incidence angles between 0° and 70°, in steps of 10° (eight angles in total). The experiment was carried out indoors at a fixed location to avoid the influences of other factors, like distance and atmosphere. The angle of 80° and the bigger ones were not set in our study due to little intensity recorded by the system. In total, we acquired 240 measurements for different samples at different angles. Each measurement was recorded three times to decrease the noise.

C. Correction Model

Bidirectional reflectance factor (BRF) is usually used to represent the reflectance of the natural target. For hyperspectral LiDAR, the reflectance is defined as the ratio of the backscattering intensity of a target to the backscattering intensity of a standard white-panel (see the following equation):

$$r_\lambda(\alpha) = \frac{I_{t,\lambda}(\alpha)}{I_{w-p,\lambda}(\alpha)} r_{w-p} \tag{1}$$

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where $r_\lambda(\alpha)$ is the reflectance of a natural target at incidence angle of α and specific wavelength of λ, $I_{t,\lambda}(\alpha)$ is the backscattering intensity of a natural target, $I_{w-p,\lambda}(\alpha)$ is the backscattering intensity of a white-panel, and r_{w-p} is the reflectance of the standard white-panel, which is a constant of 0.99 in this study.

A natural target is neither a perfect Lambertian nor a specular object, and the backscattering intensity and reflectance of the natural target surface are a combination of diffuse and specular intensity and reflectance [25]. Thus, it is necessary to build a model composed of the two parts to describe the characteristic of a natural target. Poullain *et al.* [26] proposed a linear combination of Lambertian cosine law used to represent the diffuse scattering characteristic and Beckmann law used to represent the specular characteristic to describe the backscattering intensity distribution of the coastal zone surface (see the following equation):

$$I = f\left[k_d \cos\alpha + (1-k_d)\exp\left(-\left(\tan^2\alpha\right)/m^2\right)/\cos^5\alpha\right] \quad (2)$$

where I is the backscattering intensity at a given incidence angle, f is the backscattering intensity at normal direction, k_d is the fraction of the diffuse part, α is the incidence angle, and m is the surface roughness factor.

The values of k_d and m could be estimated by the least-square method from the curve of the intensities changing with incidence angles, with the bound values are between 0 and 1. After getting the two parameters of one object, the intensities of another object of the same species at other incidence angles can be corrected to the intensity at normal direction by dividing the second term on the right side of (2). k_d and m were deemed unchanged with angles in [26], changing with the wavelengths, not related to the incidence angles.

Kaasalainen *et al.* [27] analyzed the uncertainty in multispectral LiDAR signals caused by the incidence angle effect using that model [(2)]. The limitation was that the diffuse fraction k_d and the surface roughness factor m in the model were still assumed to be related to wavelength, without considering the impact from other factors such as incidence angle. However, the surface roughness factor m is also affected by the leaf incidence angle and height difference of the leaf surface at the microscale of wavelength level. For a leaf surface to be smooth according to the following criterion [20]:

$$\Delta h < \lambda/(8\cos\alpha) \quad (3)$$

where Δh refers to the height difference between the initial contact point of the laser beam with the leaf surface at a specific incidence angle and the reflection point with laser running out the leaf surface at the microscopic scale.

Even at the same wavelength, its diffuse fractions and roughness factors are different at different incidence angles for a leaf blade. Hence, wavelength and incidence angle are supposed to be considered when representing and calculating the values of the diffuse fraction k_d and the surface roughness factor m. For hyperspectral LiDAR, (2) should be expressed as the following equation to correct the intensity data with the incidence angle effect:

$$I_{l,\lambda}(\alpha) = I_{l,\lambda}(0)\left[k_{d,\lambda}(\alpha)\cos\alpha + \left(1-k_{d,\lambda}(\alpha)\right) \times \exp\left(-\left(\tan^2\alpha\right)/m_\lambda^2(\alpha)\right)/\cos^5\alpha\right] \quad (4)$$

where $I_{l,\lambda}(\alpha)$ is the leaf backscattering intensity received by the hyperspectral LiDAR at the wavelength of λ and incidence angle of α, $I_{l,\lambda}(0)$ is the leaf backscattering intensity at the wavelength of λ and normal incidence angle, and $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ are the diffusion fraction and the surface roughness factor at the wavelength of λ and incidence angle of α, respectively, taking into consideration the wavelength and incidence angle simultaneously.

The values of $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ could be estimated by the least-square method, with the bound values are between 0 and 1. After measuring the intensities of some leaf samples at normal direction and the intensities of the same leaf samples at another incidence angle at some specific wavelength, $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ can be calculated for that wavelength of λ and that incidence angle of α. Taking the incidence angle of 10° and wavelength of 540 nm as an example, the two parameters of a leaf species need to be calculated. First, the intensities of all samples at the incidence angle of 0° are measured; then, the theoretical intensity values at the incidence angle of 10° can be expressed by multiplying the second term on the right side of (4). Moreover, the intensities of the samples at the incidence angle of 10° are also obtained in the experiment. The key point of the least-square method is to estimate two optimal values of $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ to make the difference between the measured intensity and theoretical intensity as little as possible. Finally, the values of the two parameters are estimated at the incidence angle of 10° and wavelength of 540 nm. Then, intensities at the other incidence angles and the other wavelengths are iterated to be used to participate in the calculation until the values of $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ of the tree species are obtained for all the incidence angles and all the wavelengths. Thus, the leaf diffuse fraction and surface roughness are obtained. And the backscattering intensity $I_{l,\lambda}(\alpha)$ of another leaf blade of the same species at some incidence angle of α can be corrected to the leaf backscattering intensity $I_{l,\lambda}(0)$ at the normal incidence angle by dividing the second term on the right side of (4) by selecting the corresponding $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ we got.

Furthermore, the reflectance correction model of the incidence angle effect is as (5), shown at the bottom of the next page, where $r_{\lambda,c}(\alpha)$ is the corrected reflectance of the leaf surface at the wavelength of λ and at incidence angle α, and $I_{w-p,\lambda}(0)$ is the backscattering intensity of the white-panel at normal incidence angle.

D. Leaf Incidence Angle Calculation

When it comes to the intensity and reflectance correction of the three dimensional (3-D) point cloud data of vegetation using our modified and practical algorithm [(4) and (5)] for hyperspectral LiDAR, it is essential to first calculate the incidence angle of each point of the 3-D point cloud. The incidence angle refers to the angle between the laser beam and the normal direction of the leaf surface, which is an acute angle. Thus, the normal vector of the leaf surface is required to be obtained for each point. The k-nearest neighbor (KNN) algorithm is used to search for the nearest neighbor points of each point, and it deals with boundaries and holes

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well [28]. After finding the neighbors of the current point, the parameters of the 3-D plane equation can be obtained based on the analysis of the eigenvectors and eigenvalues of a covariance matrix [29]. Besides, the 3-D plane can also be fitted by the least-square method [30]. After determining the best-fitting plane, the normal direction vector of the current point is easily obtained. Then, the incidence angle is calculated as follows:

$$\alpha = \arccos(\mathbf{L} \cdot \mathbf{N}/(\mathbf{LN})) \tag{6}$$

where α is the incidence angle, $\mathbf{L}$ is the laser beam vector, and the 3-D coordinates of the point usually express it, and $\mathbf{N}$ is the leaf surface normal vector calculated before.

Thus, the intensity of the point with incidence angle effect can be corrected to the intensity at normal direction by choosing the corresponding $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$ using the look-up table (LUT) method. The high-dimensional LUT requires dimensionality reduction and interpolation. But the LUT in this study has only the wavelength and incidence angle as 2-D variables and the wavelengths of this hyperspectral LiDAR are fixed. The LUT consists of lines, and one line content includes four items, which are the wavelength, incidence angle, and their corresponding $k_{d,\lambda}(\alpha)$ and $m_\lambda(\alpha)$. If $k_{d,\lambda}(\alpha)$ or $m_\lambda(\alpha)$ at any incidence angle is required to be estimated, the LUT is used to calculate it. First, the two nearest angles for that incidence angle are identified, and then the two lines at the wavelength and the two nearest incidence angles in the LUT are found. Finally, the inverse distance weight (IDW) method is applied to calculate the $k_{d,\lambda}(\alpha)$ or $m_\lambda(\alpha)$ at that incidence angle, where the distance refers to the difference of the incidence angles.

III. Results

Hyperspectral LiDAR measured three types of tree leaf samples and the white-panel at normal incidence angle (0°) and other incidence angles set in Section II-B, and the intensity data were acquired. Then, the intensity data were used to calculate the parameters of the model, and the correction processing was subsequently carried out.

A. Analysis of Parameters

In terms of the parameters of the diffuse fraction and the surface roughness factor, previous studies reckoned that they did not change with the incidence angles. As for the traditional single- or dual-wavelength LiDAR and multispectral LiDAR, the analysis of the incidence angle effect was based on the assumption that both the diffuse fraction and the surface roughness factor were constants at a specific wavelength. Fig. 3 shows the parameter of k_d changes with the wavelengths in the previous model before the modification. Fig. 4 shows the parameter of m changes with the wavelengths before the modification. The two parameters vary with the wavelengths

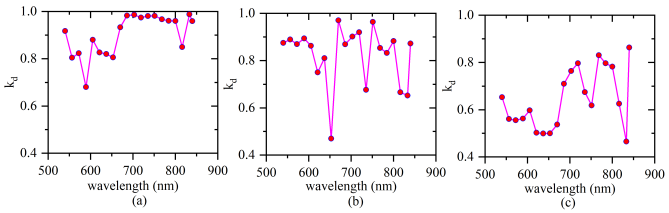


Fig. 3. Parameters of k_d of the selected three leaf samples with wavelengths. (a) *Magnolia delavayi.* (b) *Juglans regia.* (c) *Macaranga tanarius.*

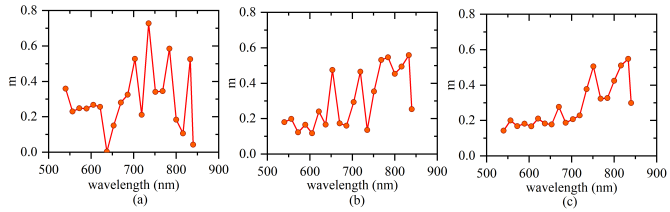


Fig. 4. Parameters of m of three leaf samples with wavelengths. (a) *Magnolia delavayi.* (b) *Juglans regia.* (c) *Macaranga tanarius.*

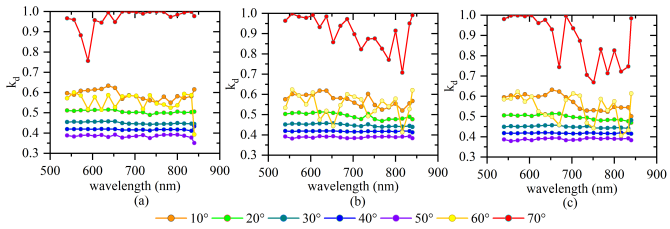


Fig. 5. Parameters of k_d leaf samples after the modification. (a) *Magnolia delavayi.* (b) *Juglans regia.* (c) *Macaranga tanarius.*

of hyperspectral LiDAR in that model, and all the incidence angles share only one curve of diffuse fraction and only one curve of surface roughness factor for a species. It means that k_d and m are constants for a given wavelength at any incidence angle. The complicated surface reflection characteristic is only associated with wavelength, unchanged with incidence angles.

However, a natural object's reflection characteristics are different at different incidence angles and not constants based on [20]. To explore the complex reflection characteristics of the leaves, we studied the relationship among the k_d and m, the wavelength, and the incidence angle. Fig. 5 shows the value of k_d changes with the incidence angle and the wavelength for the three types of leaves. When the incidence angle is lower (less than 50°), the diffuse fraction nearly keeps as a constant as the wavelength increases. When the incidence angle is 60° or 70° (bigger than 50°), the value of k_d is a function of the incidence angle and the wavelength. For a given wavelength, it can be seen that the value of k_d is different from each other at different incidence angles, which demonstrates the assumption of (4), and it exactly suggests the values of Fig. 3 need to be improved.

Fig. 6 shows the values of m as a function of incidence angles and wavelengths for the three types of leaves. For a

$$r_{\lambda,c}(\alpha) = \frac{I_{l,\lambda}(\alpha) r_{w-p}}{I_{w-p,\lambda}(0)\left[k_{d,\lambda}(\alpha)\cos\alpha + \left(1 - k_{d,\lambda}(\alpha)\right)\exp\left(-\left(\tan^2\alpha\right)/m_\lambda^2(\alpha)\right)/\cos^5\alpha\right]} \tag{5}$$

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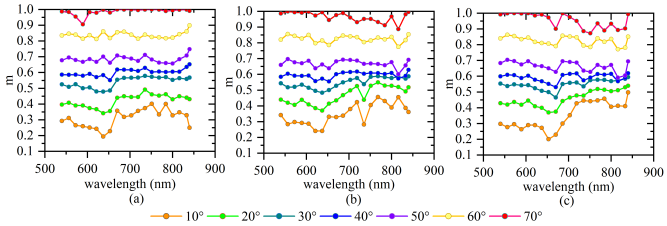


Fig. 6. Parameters of m of leaf samples after the modification. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

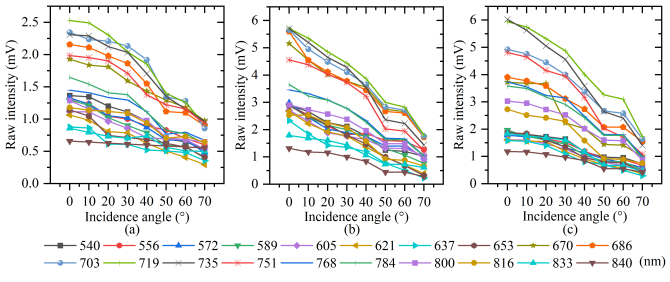


Fig. 7. Raw intensity variations of three leaf samples with incidence angles. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

given wavelength, it shows that the values of m are different from each other at different incidence angles. It should be noted that the variations of k_d and m in Figs. 5 and 6 also demonstrate the point of view in [20]. Therefore, it is appropriate for us to take into consideration the incidence angle and wavelength simultaneously to represent the complex surface reflection ability of the leaves.

B. Intensity Correction

Fig. 7 shows the raw intensity variations of three randomly selected samples (one sample a tree species) with the incidence angles. Since the instrument is a prototype, we selected the best performing 20 bands ranging from 540 to 840 nm which have a better signal-to-noise ratio (SNR) to experiment to verify our algorithm and to obtain reliable results. It can be seen that intensity decreases slowly at the incidence angles of 0°–30°, then it has a fast downward trend at 40° and 50°, and finally, the changing trend becomes smaller at the bigger incidence angles. The phenomenon is because the bigger the incidence angle is, the less backscattering intensity hyperspectral LiDAR recorded. If the incidence angle is bigger than 50°, little intensity will be recorded. Thus, the changing trend of intensity is nearly flat at the bigger incidence angles. The intensity variation displays a gentle trend with incidence angles at wavelengths shorter than 670 nm and wavelengths longer than 751 nm. The overall trend displays that the intensity decreases with the incidence angle increasing. There is a noticeable incidence angle effect within the intensity. Usually, it is mainly due to the surface roughness factor and the diffuse fraction of leaves at different incidence angles and different wavelengths.

When it comes to the quantitative applications, removing the incidence angle effect of the leaves is a must. Figs. 8 and 9 show the corrected intensities of the above three leaves with the wavelengths and incidence angles using the original model of (2) and using the proposed model of (4) by us, respectively.

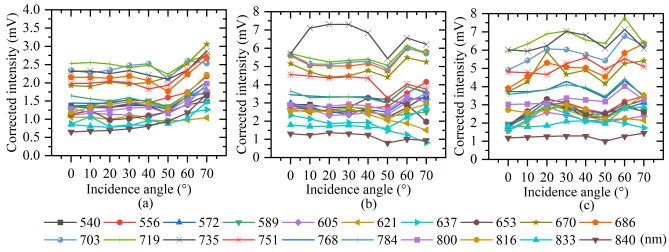


Fig. 8. Corrected intensity variations of the three leaf samples with incidence angles using previous model [see (2)]. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

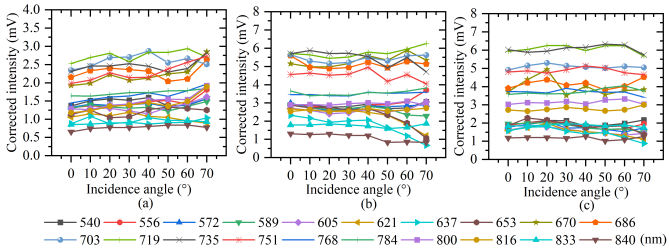


Fig. 9. Corrected intensity variations of the three leaf samples with incidence angles using our proposed model [see (4)]. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

Fig. 8 used the parameters from Figs. 3 and 4, and Fig. 9 used the parameters from Figs. 5 and 6.

Since the diffuse fraction and surface roughness factor were assumed to be associated with the wavelength in previous studies, the corrected results of intensities using (2) are not ideal, as shown in Fig. 8, where the corrected intensity of *Magnolia delavayi* displays a fluctuation at the incidence angles bigger than 40°, and so do the other two samples of *Juglans regia* and *Macaranga tanarius*. Compared with *Magnolia delavayi*, the fluctuation trend of *Juglans regia* is bigger at 735 and 686 nm. The intensities of the other wavelengths at incidence angles bigger than 50° also have a small fluctuation. Moreover, it should be noted that *Macaranga tanarius* has the most significant fluctuation with incidence angles, so the correction effect of it is quite bad. The reason for the above phenomenon can be attributed to the parameters of k_d and m, because all the intensities at different incidence angles use the same parameter curves of k_d and m shown in Figs. 3 and 4.

To compare the difference between the previous model and the modified model, the parameters of k_d and m varying with the incidence angle and wavelength in Figs. 5 and 6 were applied to the incidence angle effect correction of the intensity. The corrected results in Fig. 9 show that no significant fluctuations were found compared with the results of the previous model in Fig. 8. There are small fluctuations of the corrected intensities at 653 nm for *Juglans regia* and at 670 nm for *Macaranga tanarius*. This phenomenon can be explained by unstable systematic error when collecting intensity data, which influenced the parameters estimation and corrected results. Overall, the corrected intensities at other incidence angles are close to the intensity at normal direction. The results indicate that the model considering the incidence angle and wavelength

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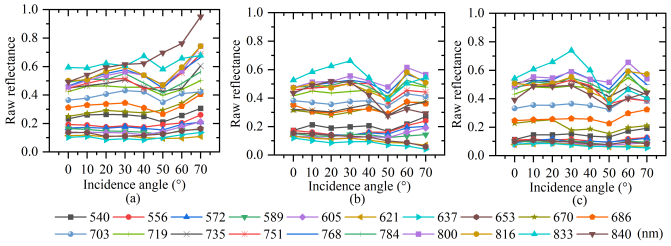


Fig. 10. Raw reflectance variations of the three leaf samples with incidence angles. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

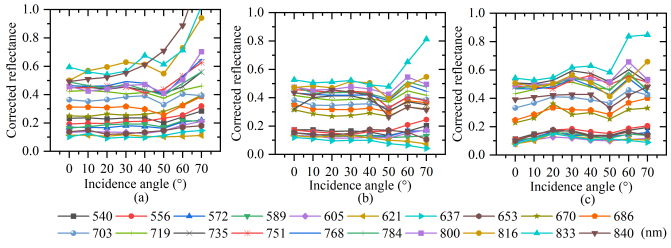


Fig. 11. Corrected reflectance variations of the three leaf samples with incidence angles using the previous parameters in Figs. 3 and 4. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

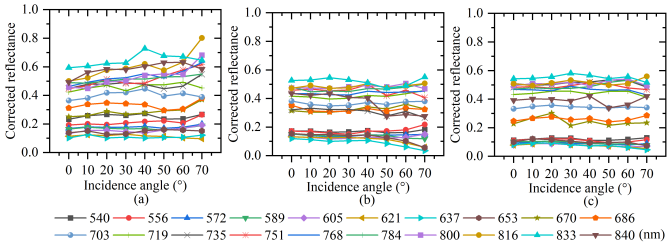


Fig. 12. Corrected reflectance variations of the three leaf samples with incidence angles using our proposed model [see (5)]. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

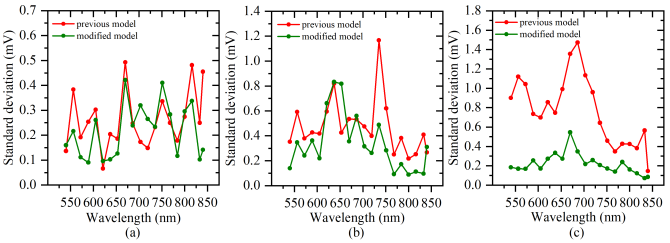


Fig. 13. Standard deviations of corrected intensities of the three leaf samples using the previous model and modified model. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

together is better than the previous model and the overall incidence angle effect disappeared using that modified model.

C. Reflectance Correction

Intensity cannot represent the real reflection characteristic of the object. The proper means of representing an object's reflection ability is the reflectance, of which the calculation formula is (1). The reflectance of the above three leaf samples was shown in Fig. 10.

It shows that the reflectance has a gentle trend at the incidence angles of 0°–40° at most of the wavelengths for the three tree leaf samples, except at the wavelength of 840 nm for *Magnolia delavayi* and 833 and 840 nm for *Juglans regia* and *Macaranga tanarius* at the incidence angles of 0° to 30°. However, reflectance at the wavelengths longer than 703 nm at 50°–70° has a significant increment for *Magnolia delavayi* and has an upward and downward trend for *Juglans regia* and *Macaranga tanarius*. It reveals that reflectance of leaves varies with incidence angle and wavelength.

The incidence angle effect will cause errors if not corrected when the directional reflectance is applied to carry out the vertical biochemical parameter estimation. Therefore, it is necessary to remove the effect before subsequent and quantitative applications. Figs. 11 and 12 show the corrected reflectance of the above three leaves with the wavelength and incidence angles changing using the original model and the modified (5) proposed by us, respectively. Fig. 11 used the parameters from Figs. 3 and 4, and Fig. 12 used the modified parameters from Figs. 5 and 6.

As shown in Fig. 11, the corrected reflectance has a great increment, which is about to be close to 1 at the wavelengths of 816, 833, and 840 nm at 40°–70° of *Magnolia delavayi*. It is apparent from Fig. 8 that this phenomenon is due to the significantly increased intensities after correction using the parameters of the previous model. The reflectance of *Juglans regia* and *Macaranga tanarius* at 833 nm at the incidence angles bigger than 50° has a significant increment, and it is also due to the increased intensities using the previous model in Fig. 8. Besides, the reflectance of *Juglans regia* at 833 nm at the incidence angles of 0°–50° has a large fluctuation. It should be noted that the corrected results of the previous model still do not have a flat trend with incidence angles and are not close to the reflectance at normal direction at most of the wavelengths.

To compare the difference between the previous model and the modified model, the parameters of k_d and m varying with the incidence angle and wavelength in Figs. 5 and 6 were applied to the incidence angle effect correction of the reflectance. Compared with the corrected results of the previous model, the corrected reflectance in Fig. 12 with the modified model shows no significant fluctuations, except a small increasing trend at 816 nm at 70° for *Magnolia delavayi* and a small decreasing trend at 840 nm at 50° for *Juglans regia*. Overall, the corrected reflectance at other incidence angles is close to the reflectance at normal direction.

D. Evaluation of Correction Effect

To further quantitatively assess the correction effect, the standard deviation was chosen to evaluate the performances of the models. The use of models is to correct the intensities and reflectance at other incidence angles to the intensity and reflectance at normal direction. Thus, the intensity and reflectance at the incidence angle of 0° are the standard values, and the standard deviations of the intensities and reflectance at other incidence angles for each wavelength are calculated. Fig. 13 shows the standard deviations of corrected intensities using the previous model and our modified model. Fig. 14 shows the standard deviations of corrected reflectance using the previous model and our modified model.

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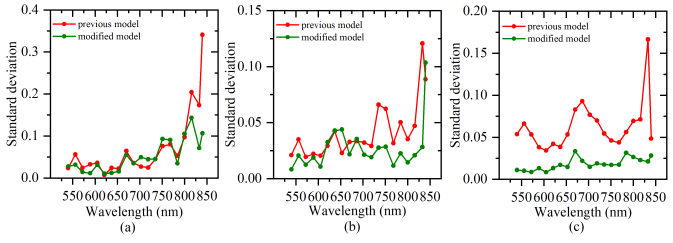


Fig. 14. Standard deviations of corrected reflectance of the three leaf samples using the previous model and modified model. (a) *Magnolia delavayi.* (b) *Juglans regia*. (c) *Macaranga tanarius.*

As can be seen from Fig. 13, the standard deviations of corrected intensities with the modified model are less than the previous model at most of the wavelengths. The performance of the modified model is totally better than the previous model for *Macaranga tanarius* especially. The biggest standard deviation of corrected intensities using the modified model is 0.83 for the above samples, which is still less than the value of 1.47 using the previous model. From Fig. 14, it can also be noted that the standard deviations of corrected reflectance using the modified model are generally less than that of the previous model. The biggest standard deviation of the previous model occurs in *Magnolia delavayi* at the wavelength of 840 nm, which is 0.34, but the biggest standard deviation of the modified model is 0.14, still less than that of the previous model. The reflectance correction effect using the modified model is better than the previous model.

We can conclude from the above results that the model taking into consideration the incidence angle and wavelength together is better than the previous model considering the wavelength, and the overall incidence angle effect disappeared using the modified model. The results clearly show that the modified models achieve excellent performance.

Therefore, it can be concluded from the intensity and reflectance changes before and after the correction that the modified correction models can be applied in radiometric correction for incidence angle effect, considering the impact of the wavelength and incidence angle on the diffuse fraction and the roughness factor simultaneously.

IV. Conclusion

The incidence angle effect is a key point that needs to be removed when using reflectance with the incidence angle effect to estimate the vertical biochemical parameters of vegetation and carry out other quantitative applications. Previous studies reckoned that the diffuse fraction and surface roughness factor are related to the wavelength and neglected the impact of the incidence angle. According to [20], the roughness factor is related to the wavelength, incidence angle, and height difference of leaf surface at the microscale of wavelength level. It is so hard and laborious to measure the height difference of the leaf surface that we do not consider it in the modified model. Hence, we take into consideration the impact of wavelength and incidence angle together in our algorithm. The proper correction model of intensity data recorded by hyperspectral LiDAR is introduced, and the reflectance correction model of hyperspectral LiDAR is also proposed. Leaves collected from three tree species were used to participate in the experiment and verify the models' performances. The main contributions and conclusions can be summarized as follows:

1) The values of diffuse fraction and surface roughness are different from each other at different incidence angles, and a single diffuse fraction and surface roughness at some wavelength cannot be shared for all the incidence angles.
2) As shown in Fig. 9, the intensity correction model can correct the intensities at other incidence angles to the intensity at normal direction for most wavelengths. It could get better correction results than the previous model.
3) As shown in Fig. 12, the reflectance correction model also displays an outstanding performance, correcting the reflectance at other incidence angles to the reflectance at normal direction. It could also get better correction results than the previous model.
4) The standard deviations of corrected intensities and reflectance with the modified model are less than the previous model at most of the wavelengths.
5) It is a novel and physically based correction algorithm for the incidence angle effect of hyperspectral LiDAR besides the laser ratio indexes and empirical fitting. It supplies the hyperspectral LiDAR with a solution for solving the complex reflection characteristic of the object surface.

In summary, the intensity and reflectance correction models achieved good performances. They could promisingly be fundamental and thorough solutions to remove the incidence angle effect for the novel hyperspectral LiDAR due to being physically based. However, the instrument has not yet been carried on an aviation platform, so the canopy experiment still needs to be studied. Future work will concentrate on the aviation experiment and explore to do the canopy intensity and reflectance corrections.

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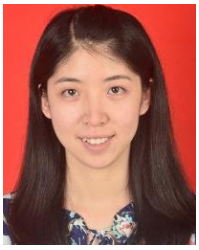
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